

BRSM

Reliability & Outliers

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Reliability



- **consistency** and **stability** of a research instrument (ex: measure or score or person)
- any measure we use in research should be reliable, otherwise it's useless
- **repeatability** of a method/test or research findings

Kinds of Reliability

- Tools/methods or measuring device



- People



Threats to Reliability

- **measurement error**: equipment malfunction, human error, or ambiguous wording in survey questions
- **instrumentation changes**: measurement instruments are not consistent across repeated measurements, changes in the instrument itself can introduce variability and affect reliability.
- **practice effects**: Participants might improve their performance in a task due to practice or learning effects, leading to different results on subsequent administrations
- **sampling variability**: In experiments involving small sample sizes, random fluctuations in the characteristics of the participants can lead to unreliable results.

Threats to Reliability

- **participant error**: any factor which adversely alters the way in which the participant responds
 - ex: interview at 11 am vs 6 pm
- **participant bias**: any factor which produces a false/biased response
 - ex: mental health questionnaire in a company
- **researcher error**: any factor which alters the researcher's interpretation
 - ex: fatigue effects if interview all day
- **researcher bias**: any factor which induces bias in the researcher's recording of responses
 - ex: subjective interpretation (to get the “result” you expect)

Kinds of Reliability

stability and **degree of agreement**
between **people** during measurements

stability and **consistency** of
method/tool/apparatus
over time/repeated
measurements

Intra-Rater
Inter-Rater
Reliability

Test-Retest
Reliability

Internal
Consistency

Parallel
Alternate
Form

coherence of attributes constituting the
method/tool/apparatus

equivalence of two versions of
the method/tool/apparatus to
compare results

Kinds of Reliability

Cohen's Kappa (nominal; 2 raters)
Fleiss' Kappa (nominal; >2 raters)
Kendall's coefficient of concordance (ordinal)
Krippendorff's Alpha (all measurement levels)

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Cronbach Alpha
Split-Half
Kuder Richardson-20/21

Pearson's correlation



Reliability

- For people (reliability of participants)
 - Inter-rater or Inter-observer Reliability - degree of agreement between two participants or observers simultaneous recorded measurements
 - correlation, helps in outlier detection
 - Intra-observer Reliability - degree of agreement within the same observer's measurements on repeated occasions



Reliability

EXAMPLE

- For people (reliability of participants)
 - Inter-rater or Inter-observer Reliability

Does this specimen have a chin?

Chimpanzee Pan troglodytes				Kevin	Mayla
Australopithecus afarensis				1. No	1. No
Paranthropus boisei				2. No	2. No
Homo erectus				3. No	3. Yes
Homo sapiens				4. No	4. No
				5. Yes	5. Yes

<http://www.passbiology.co.nz/biology-level-3/human-evolution>

1, 2, and 4 probably don't; 5 probably does.

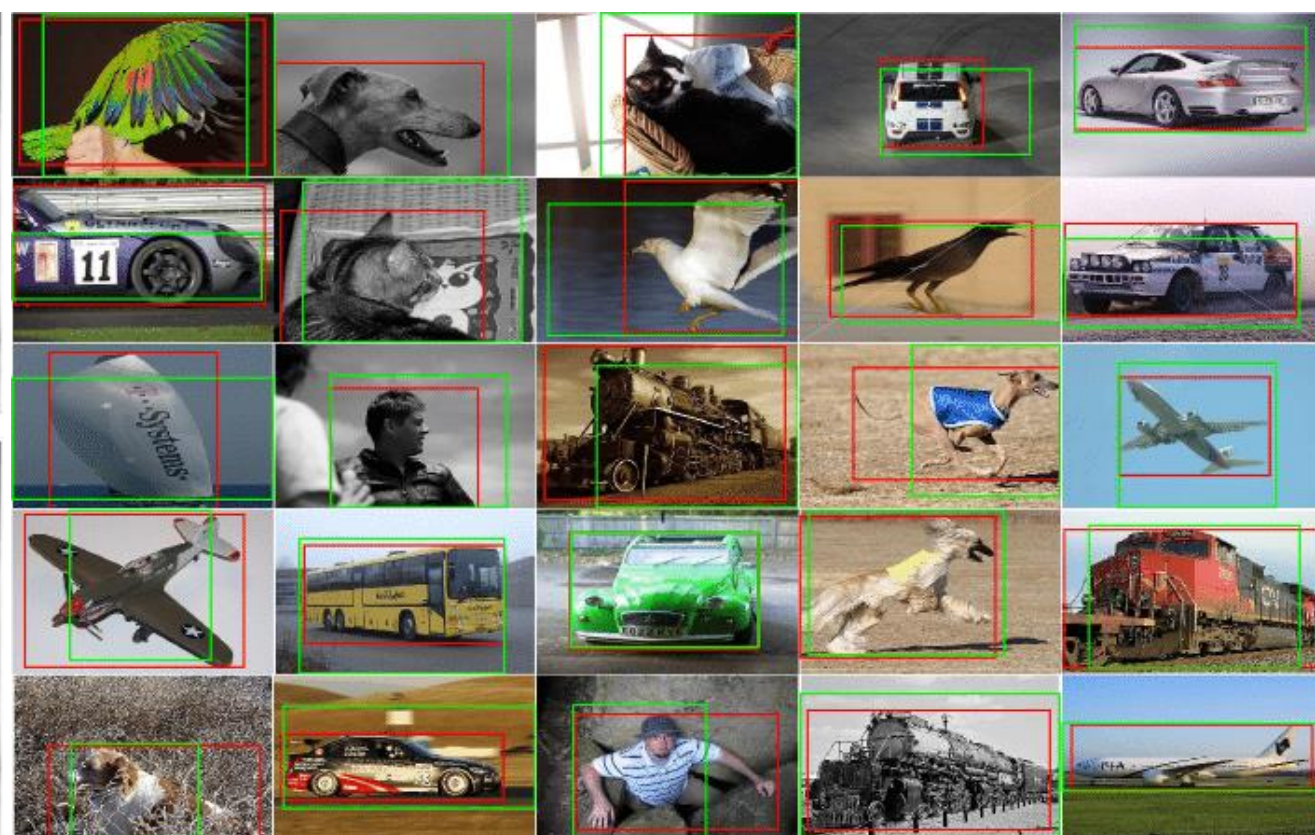
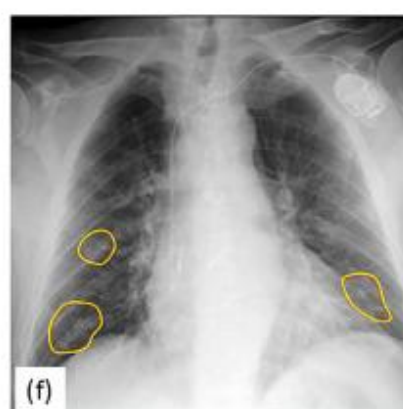
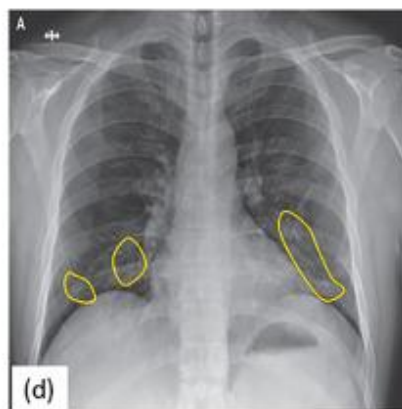
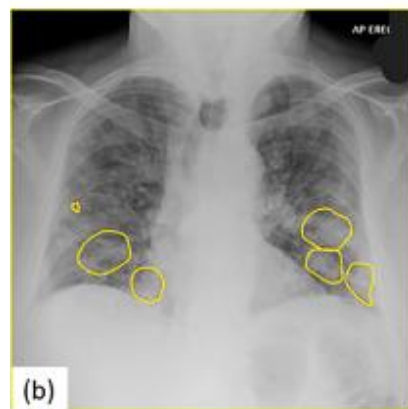
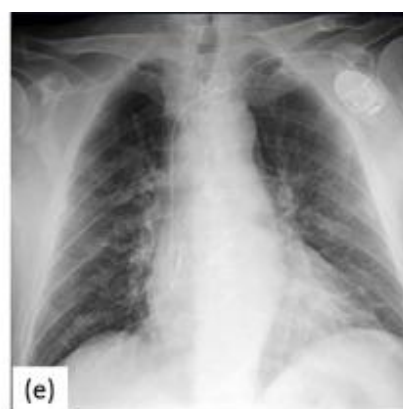
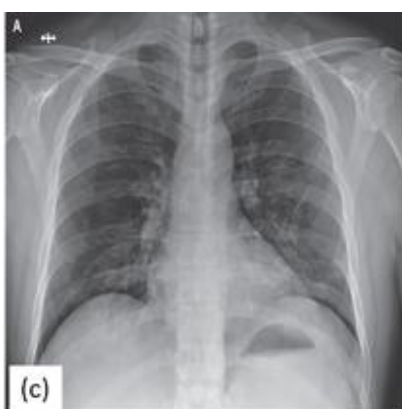
https://www.youtube.com/watch?v=fq_LNTPgVF8&app=desktop



Reliability

EXAMPLE

- For people (reliability of participants)
 - Inter-rater or Inter-observer Reliability



How many annotators per dataset?



Reliability

- ▶ *Cohen's kappa*: a quantitative measure of reliability for two raters that are rating the same thing, correcting for how often the raters may agree by chance
- ▶ can be used to check consistency of the same rater at two different time points
- ▶ used when the variable is nominal

		Rater 2	
		Correct	Incorrect
Rater 1	Correct	A	B
	Incorrect	C	D

50 images rated by 2 raters

	Yes2	No2
Yes1	4	16
No1	15	15

Reliability

- ▶ *Cohen's kappa*: a quantitative measure of reliability for two raters that are rating the same thing, correcting for how often the raters may agree by chance

```
r1=['yes','no','yes','no','yes','no','yes','no','yes']
```

```
r2=['yes','yes','yes','no','no','no','yes','yes','yes']
```

Agreement= sum of agreements /
total number of instances = $(4+2)/9 = 0.66$

	Yes2	No2
Yes1	4	1
No1	2	2

Kinds of Reliability

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Reliability

- Internal consistency: Is the measurement device consistently measuring what you want it to measure?
 - Average inter-item correlation finds the average of all correlations between pairs of questions
 - Split Half Reliability: all items that measure the same thing are randomly split into two. The two halves of the test are given to a group of people and find the correlation between the two. The split-half reliability is the correlation between the two sets of scores.
 - Kuder-Richardson 20: average correlation for all the possible split half combinations in a test.



Reliability

- Internal consistency: Is the measurement device consistently measuring what you want it to measure?
 - *Cronbach's alpha*:
 - was developed in 1951 by Cronbach Lee to meet the need of finding an objective way of measuring the internal consistency reliability of an instrument used in a research work
 - mostly used when the research being carried out has multiple-item measures of a concept
 - typically used in questionnaires/surveys (self-reported)



- Internal consistency: Is the measurement device consistently measuring what you want it to measure?
 - *Cronbach's alpha*:

- ▶ $\bar{r}$ = mean inter-indicator correlation
- ▶ k = number of indicators or number of items



EXAMPLE

0 = Never 1 = Almost Never 2 = Sometimes 3 = Fairly Often 4 = Very Often



Reliability



- Internal consistency:
 - we have a 5 item scale showing data collected from 100 respondents
 - Correlate 100 responses x 5 items matrix

	Item 1	Item 2	Item 3	Item 4	Item 5
Item 1	1.0				
Item 2	.35	1.0			
Item 3	.42	.31	1.0		
Item 4	.25	.38	.41	1.0	
Item 5	.21	.36	.46	.31	1.0

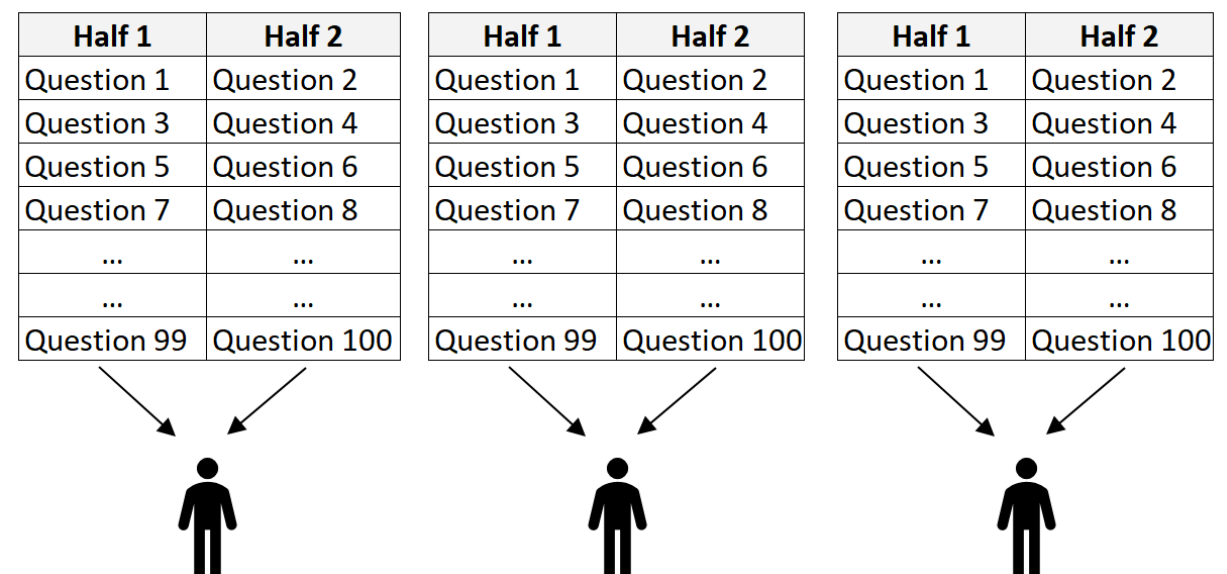
$$\alpha = \frac{k\bar{r}}{(1+(k-1)\bar{r})} = .73$$

Cronbach's alpha	Internal consistency
$\alpha \geq 0.9$	Excellent
$0.9 > \alpha \geq 0.8$	Good
$0.8 > \alpha \geq 0.7$	Acceptable
$0.7 > \alpha \geq 0.6$	Questionable
$0.6 > \alpha \geq 0.5$	Poor
$0.5 > \alpha$	Unacceptable



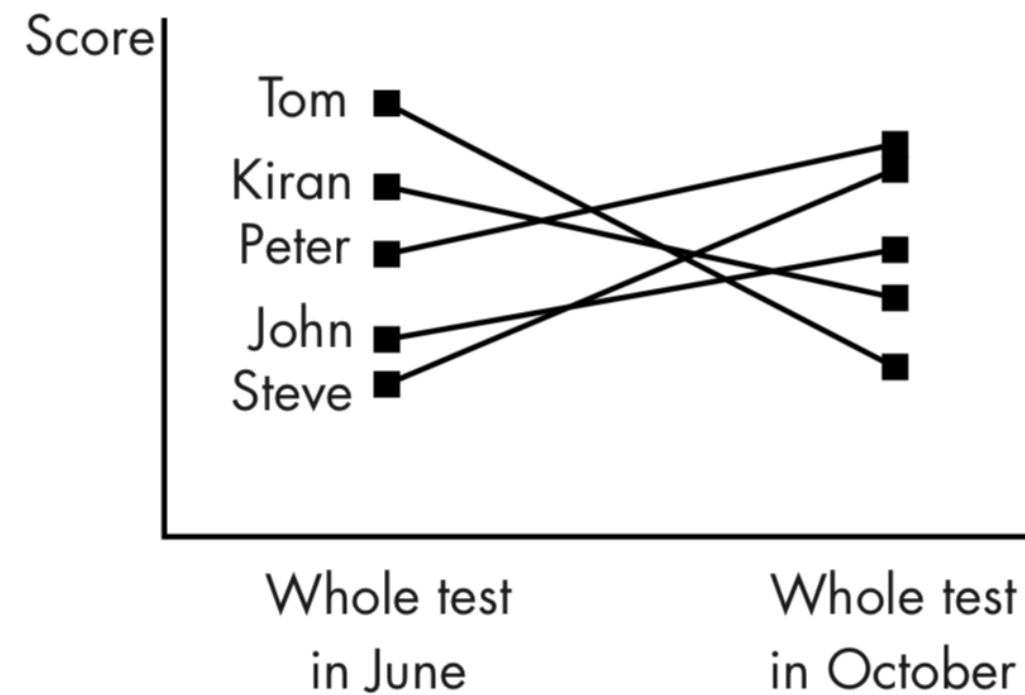
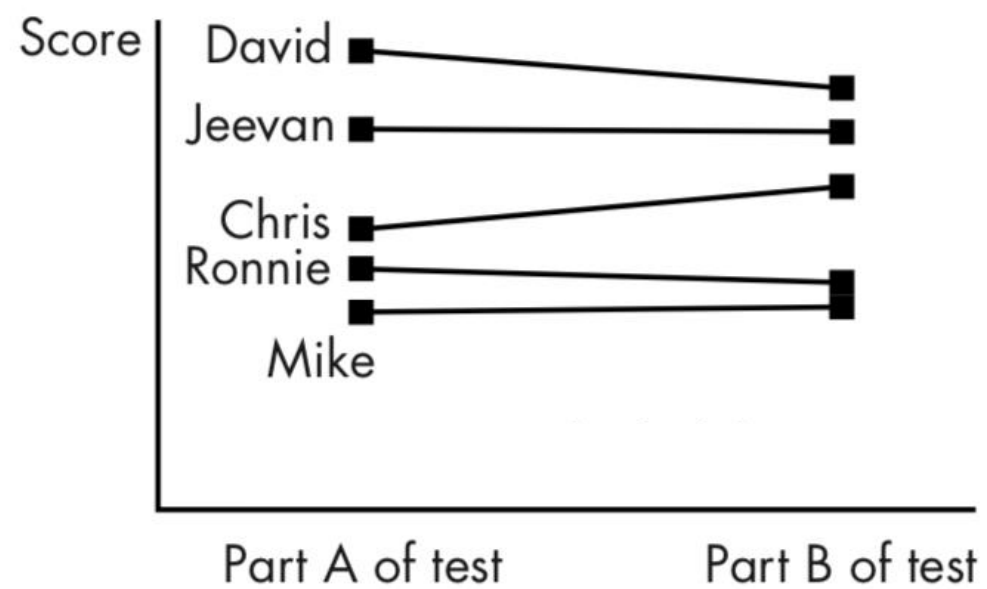
Reliability

- Internal consistency: Is the measurement device consistently measuring what you want it to measure?
 - *Split-half*:
 - uses only some of available correlations;
 - compare results of one half to the other half.
 - If the test is reliable then people's scores on each half should be similar





Reliability

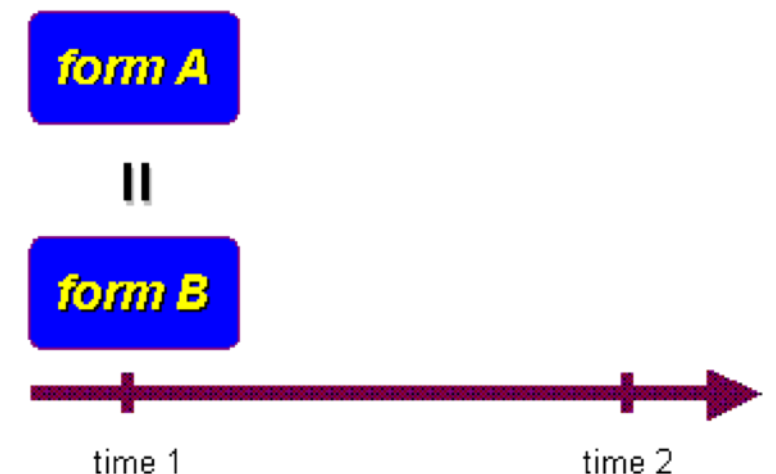


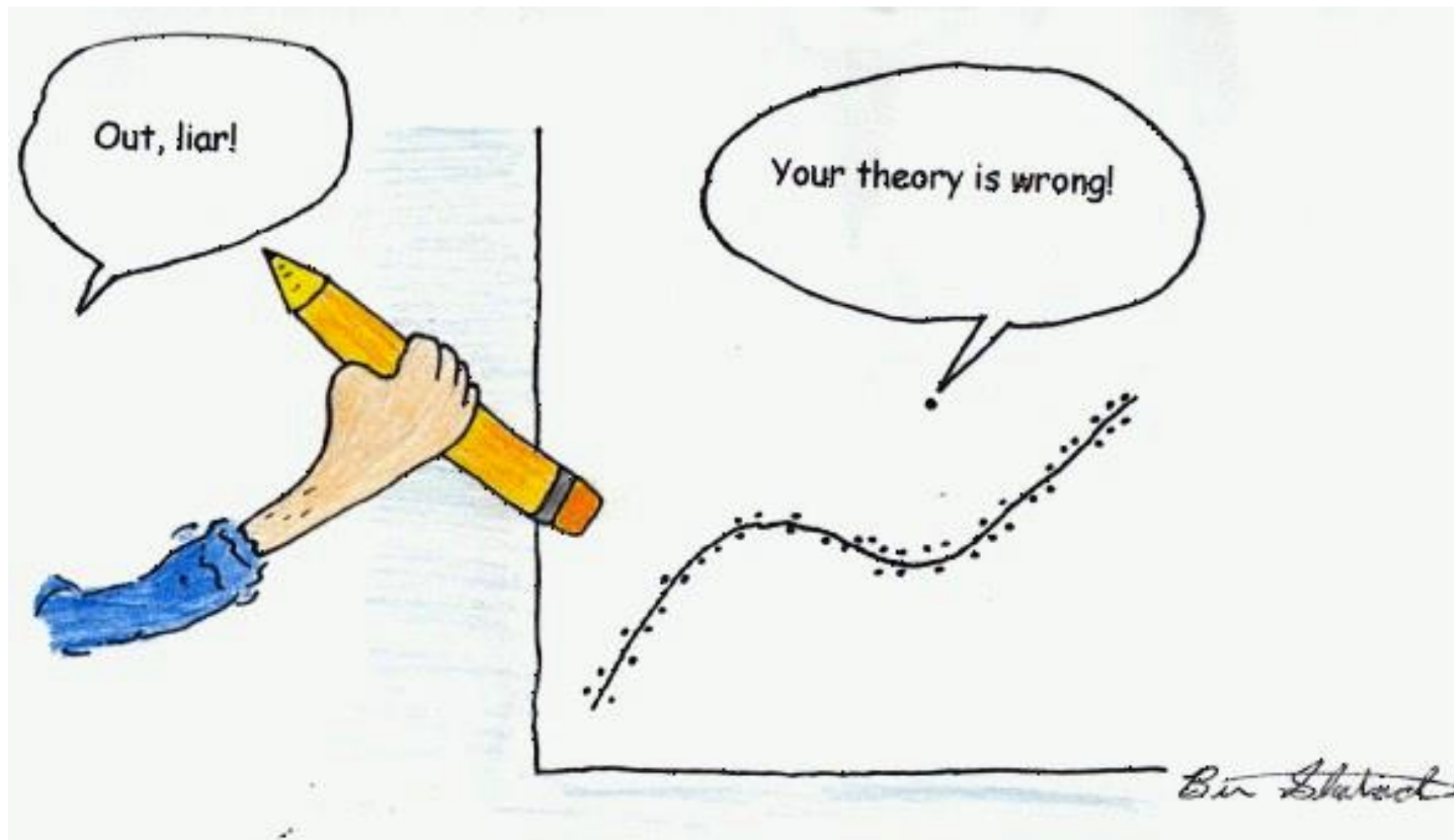
What kind of reliability and how good/bad is it?



Reliability

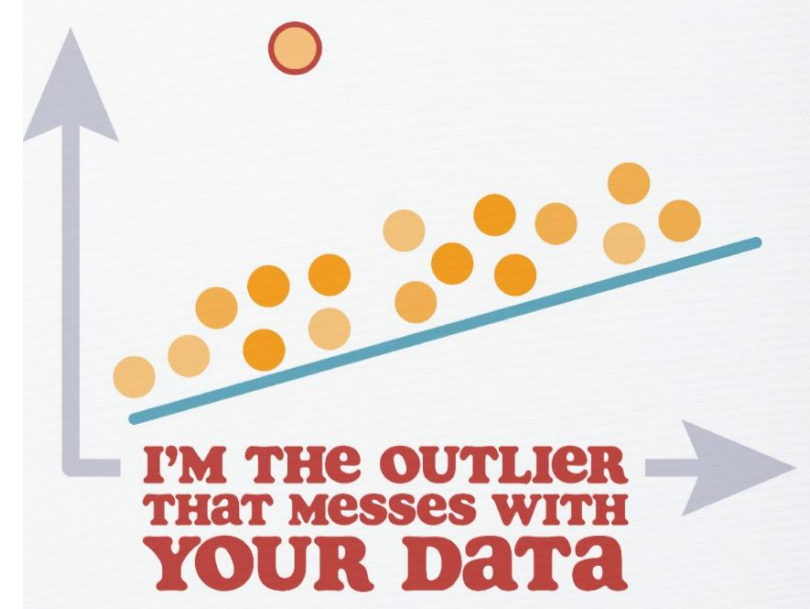
- parallel forms:
 - measure of reliability obtained by administering different versions of an assessment tool (both versions must contain items that probe the same construct, skill, knowledge base, etc.) to the same group of individuals
 - can avoid some problems inherent with test-retesting





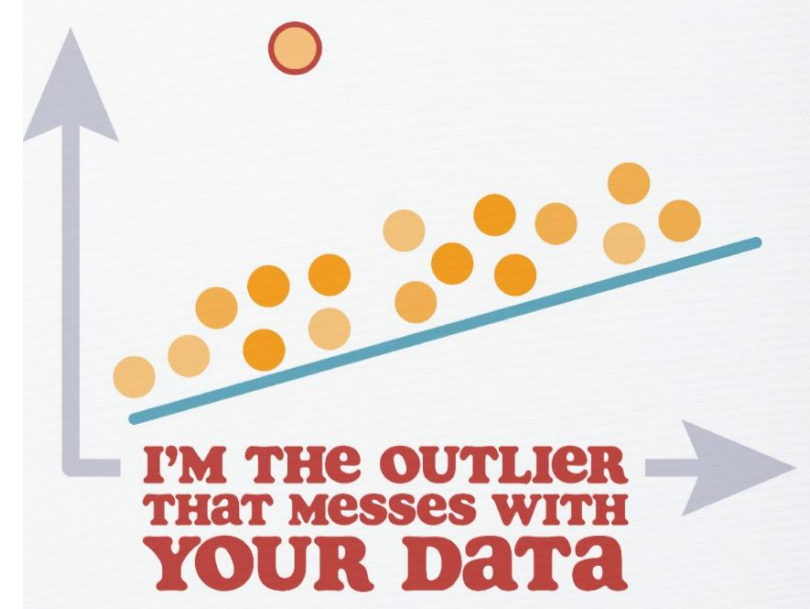
To have or not to have

Outliers



- detecting outliers is of major importance for almost any quantitative discipline (ie: Physics, Economy, Finance, Machine Learning, Cyber Security, Cognitive Science)
- not as common when sample size is low
 - ex: neuroimaging, qualitative studies involving interviews
- individual vs item/scale/stimulus

Outliers

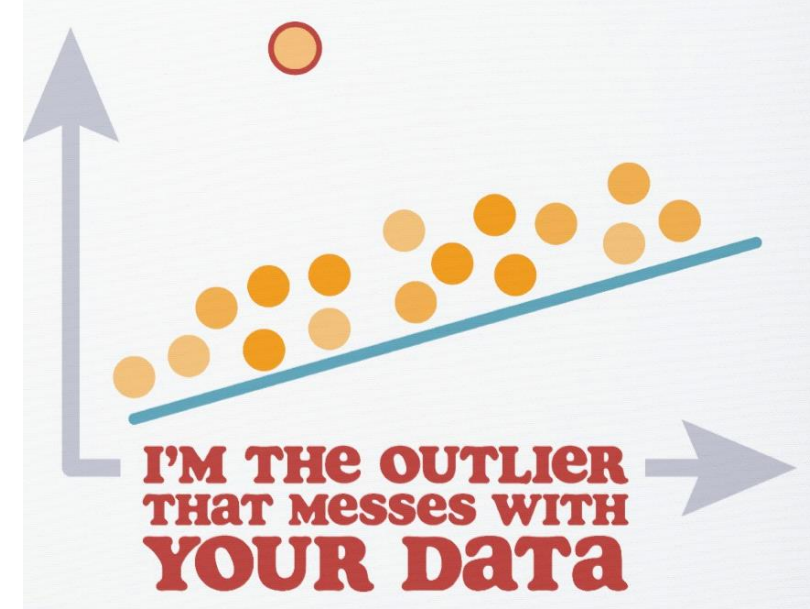


- probable causes?
 - measurement/execution errors (instrument errors/data extraction or experiment planning errors)
 - eg: improper scanner handling
 - data entry errors, missing data (human errors)
 - eg: entering 999 for missing values and using it for analysis

Dealing with Outliers

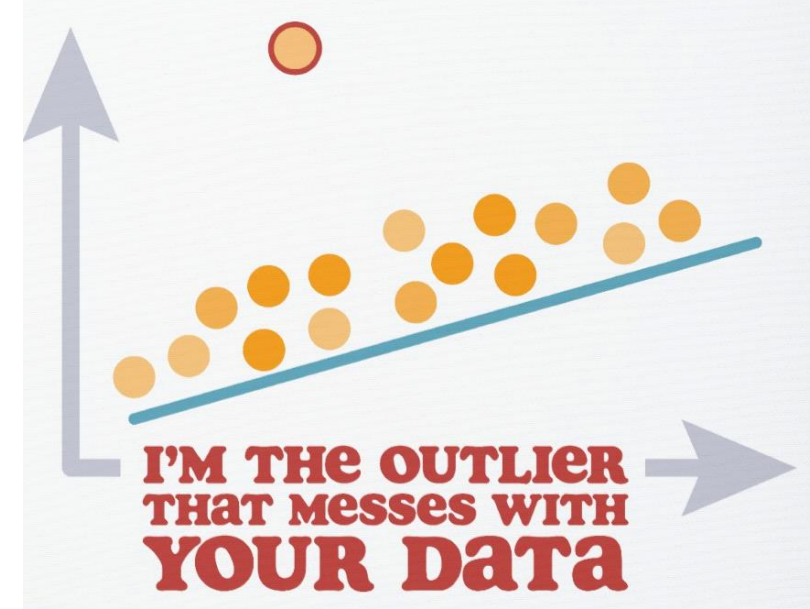
- omit
- replace (ex: with mean)
- using different analysis methods (ex: non-parametric tests)
- valuing the outliers
- data transformation

Outliers



- probable causes?
 - measurement/execution errors (instrument errors/data extraction or experiment planning errors)
 - eg: improper scanner handling
 - data entry errors, missing data (human errors)
 - eg: entering 999 for missing values and using it for analysis
 - data processing errors (data manipulation or data set unintended mutations)
 - eg: multiplying interval data

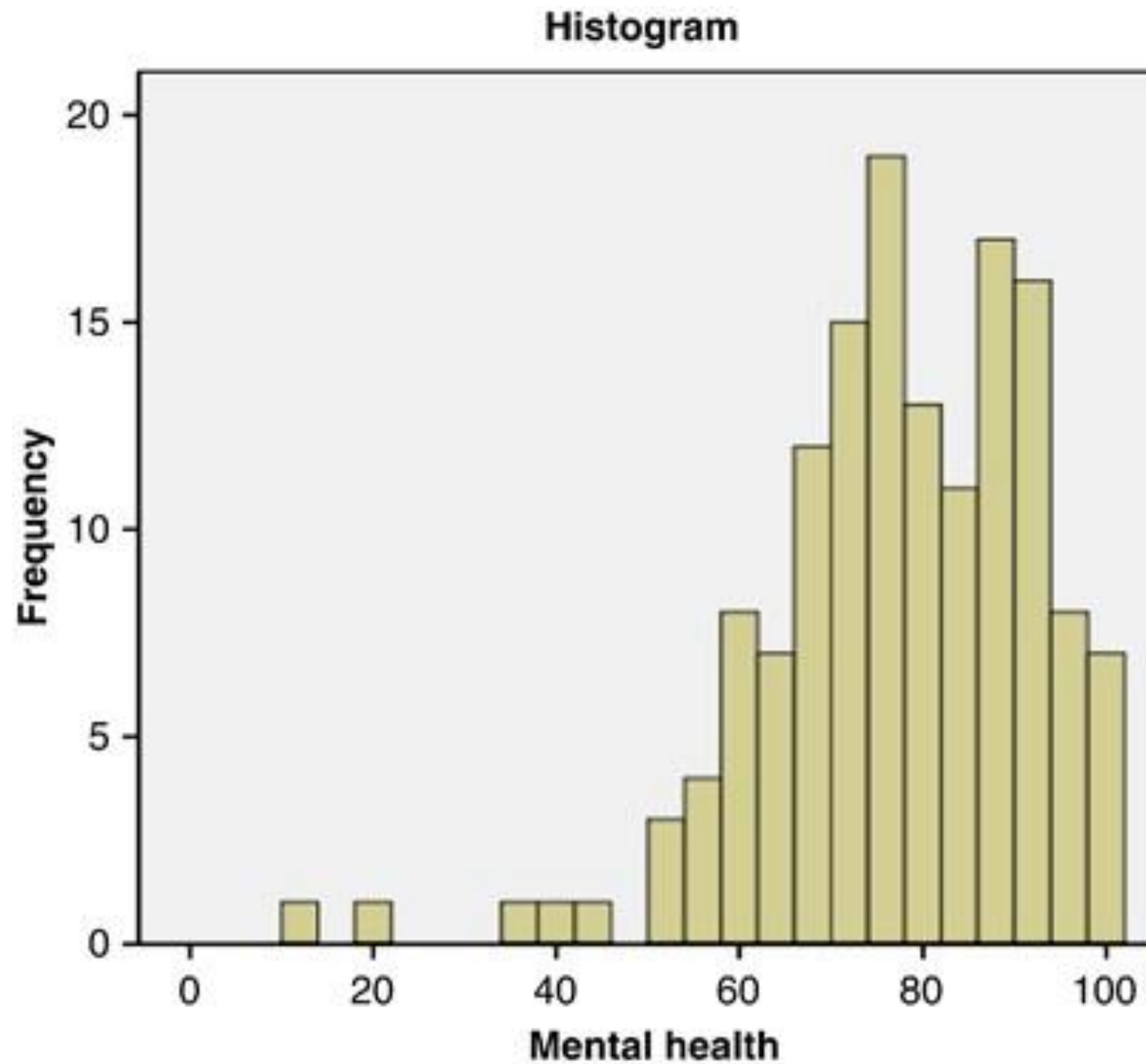
Outliers



- probable causes?
 - sampling errors (extracting or mixing data from wrong or various sources)
 - e.g: measure the weight of athletes but also include some wrestlers
 - natural (not an error, novelties in data or inherent data variability)

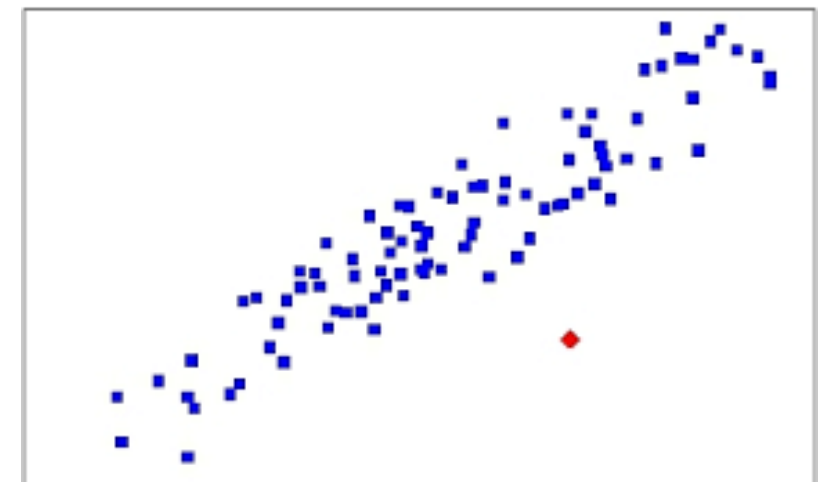
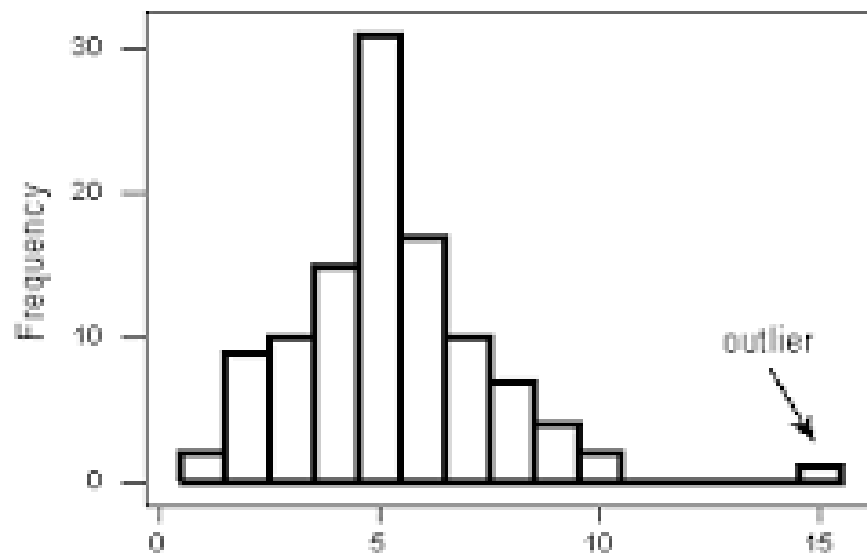
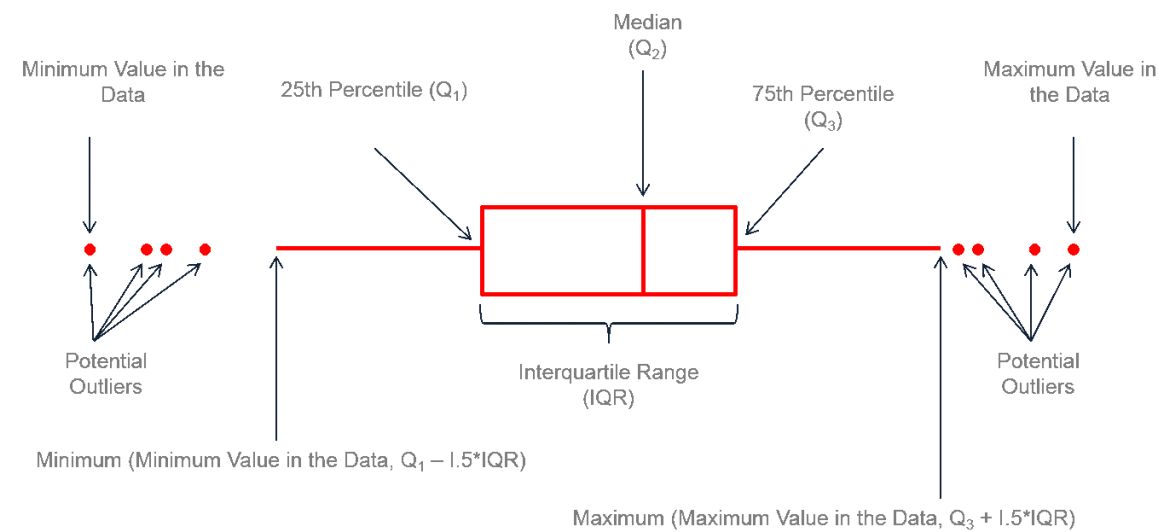
EXAMPLE

Natural Outliers



Outlier Detection

- graphical representations help (eg: scatter plot, box plot, histogram)



Outlier Detection

- graphical representations help (scatter plot, box plot, histogram)
- $>1.5 \times \text{InterQuartile Range}$
- $2/3$ SDs from mean (depending on the nature of data)
- Grubbs' test (single), Tietjen-Moore test (multiple), etc..